

CV - FARHAD KIANIFARD

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SUMMARY

Physics-Informed AI Specialist and Serial Tech Entrepreneur with over 25 years of experience leading R&D, AI product development, and high-tech startups. Pioneers the integration of Physics-Informed Neural Networks (PINNs) and Digital Twins for Big Pharma and Advanced Materials, reducing 5–7 years of wet-lab formulation trials to hours of validated simulation. Combines deep algorithmic knowledge with strong business acumen and proven executive leadership across AI, cybersecurity, and geospatial software.

CORE COMPETENCIES

- **AI & ML:** Deep Learning, Physics-Informed Neural Networks (PINNs), PyTorch, Model Optimization & Convergence.
- **Simulation & Physics:** Multiphysics Modeling, Release Kinetics (Zero-Order), Fickian Diffusion, Rheology, Polymer Degradation, Fluid Dynamics.
- **Data & Software Engineering:** Python, MATLAB, SQL, Data Preprocessing, Git.
- **Management & Strategy:** Executive Leadership, R&D Strategy, Product Lifecycle Management, Cross-functional Team Leadership, B2B Consultancy.
- **Domain Expertise:** Sustained-Release Drug Depots, In-Situ Forming Polymers, Biopharmaceuticals, Self-Healing Smart Hydrogels, AI-driven Geospatial Software.

PROFESSIONAL EXPERIENCE

CEO & Founder | Abanagah Rayaneh Sarir (BaniBal) | *2016 – 2022*

Founded a software company specializing in AI-driven geospatial services, developing the flagship product "BaniBal" (BaniBal Smart Router).

- Architected the core AI engine for the BaniBal Android navigation software, combining real-time traffic and logistics data with machine learning algorithms for smart route optimization.
- Launched the product in 2017, achieving a high user rating (4.5/5 stars) on major app platforms and successfully deploying it for large-scale enterprise clients.
- Managed the complete product lifecycle from conceptualization to mass-market deployment, leading both technical and business teams.

CEO & Founder | SafaPars | *2013 – 2016*

Founded a specialized instrumentation and solutions company providing custom data analytics and industrial automation frameworks for large industrial and governmental organizations.

- Led high-level technical consultancy and solution design for complex industrial data processing use-cases, bridging the gap between raw sensor data and actionable business intelligence.
- Oversaw multidisciplinary engineering teams, ensuring the delivery of robust, tailored software architectures for diverse enterprise clients.

CEO & Founder | Ajin | *2010 – 2013*

Established a pioneering cybersecurity firm focused on computer systems protection and network/data integrity.

- Designed and implemented high-security network architectures and enterprise threat mitigation strategies for corporate clients.
- Built a reputation for reliability in safeguarding highly sensitive organizational data during the early era of corporate cybersecurity.

INDEPENDENT AI CONSULTANCY (ADVANCED R&D PROJECTS)

Executed high-value, proprietary R&D contracts for leading pharmaceutical and advanced materials corporations, leveraging proprietary Digital Twin frameworks.

1. Senior AI Simulation Specialist | [BaniTwin](#) Project (Big Pharma)

Engineered a 30-Day In-Situ Forming Depot Digital Twin for **Tirzepatide** (Metabolic/Diabetes).

- **Impact:** Built a PINNs-based simulation reducing R&D timelines from years to hours. Validated a 30-day zero-order release profile with <4% initial burst, ensuring patient safety and maintaining therapeutic serum concentration for 30+ days.

2. Senior AI Simulation Specialist | [BaniFlex](#) Project (Big Pharma)

Designed a sustained intra-articular delivery Digital Twin for **Lorecivivint** (Osteoarthritis).

- **Impact:** Implemented electrostatic "Stealth Shielding" and active pH buffering. Achieved an 85.5% validated simulation probability, maintaining joint residence >30 days and a stable microclimate pH (~6.76).

3. Senior AI Simulation Specialist | [BaniNex](#) Project (Big Pharma)

Developed a Liquid Crystal Phase-Transition simulator for **Risperidone** (Schizophrenia).

- **Impact:** Designed interactive control systems and PyTorch-based PINN solver. Achieved an exceptional physics-constrained model loss of 0.0002 over 8,000 iterations, ensuring precise 30-day zero-order release kinetics.

4. Senior AI Simulation Specialist | [BaniShield](#) Project (Leading Advanced Materials Co.)

Designed the Digital Twin for an **ultra-hydrophobic, self-healing smart material** at room temperature.

- **Impact:** Replaced expensive physical stress-testing with a validated digital roadmap, allowing the manufacturer to optimize production processes and material durability.

EDUCATION

M.Sc. in Computer Science | Najafabad University | *2003*

Thesis: "Simulation of a Breathable Ultra-Hydrophobic Material for Enhancing Building Facade Longevity"

CERTIFICATIONS

- **Certificate of Completion: Applied Artificial Intelligence** — Karaj Azad University (*Score: 100/100*)

LANGUAGES

- **Persian:** Native
- **English:** Professional proficiency in reading and technical writing; Intermediate speaking.